

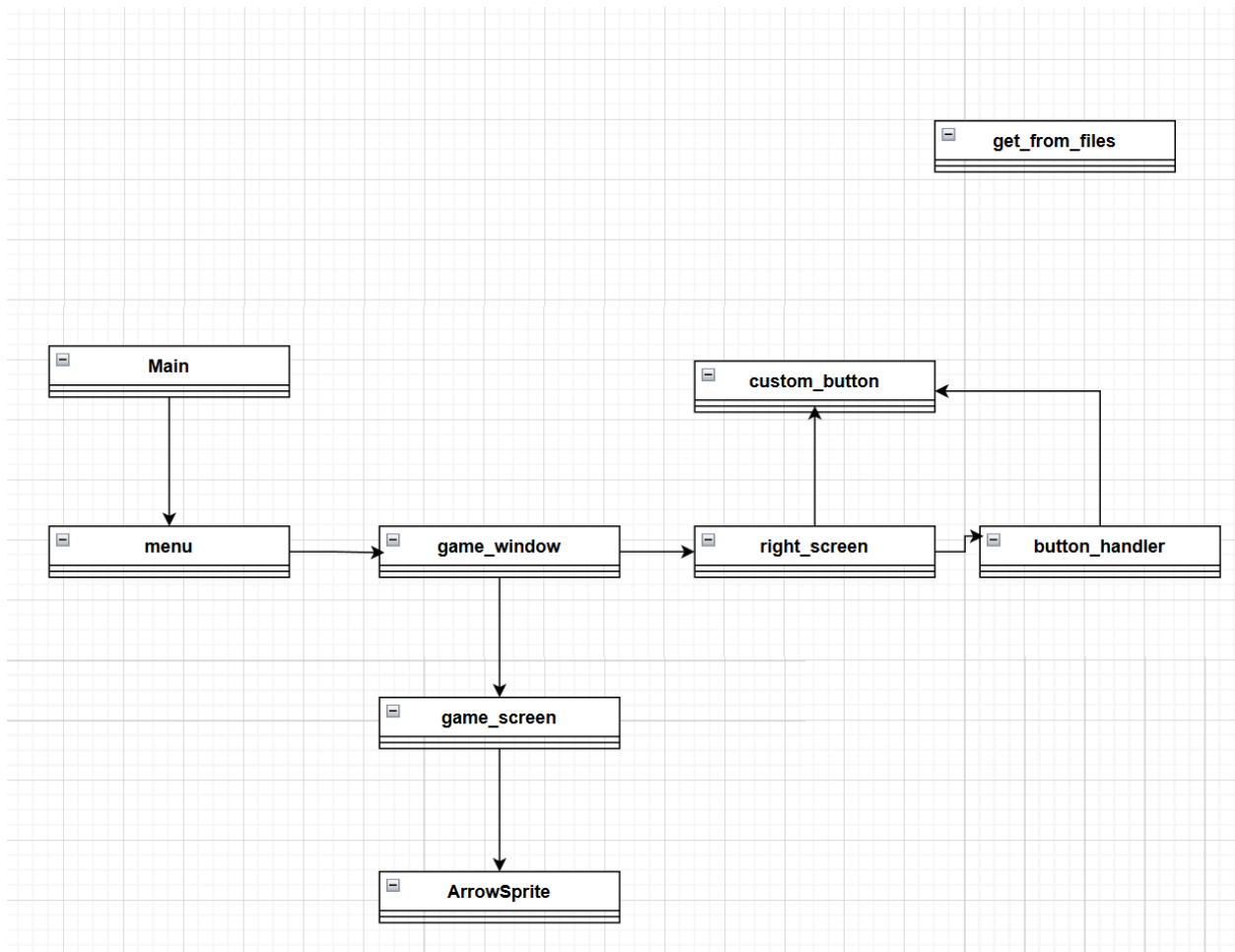
Final Project: CS 2450

Adam Carling

CS 2450

Currently, The system will output an arrow to the screen that scrolls down the screen. The player can press the corresponding keypad button and make the arrow disappear. The system is highly modular and customizable. The styles and config sheet can be changed to edit the general feel of the game without the need to edit any code.

The current design structure is as follows:



We used a highly modular design pattern. Every class is in its own file and each class is only in charge of handling a few things. This allows for design changes and easy fixes to potential mistakes during design. Like possibly thinking tkinter can be used to make a game and trying to use it for everything but realizing three classes in that it won't work. Main

and Menu classes are Tkinter windows but after that its all pygame-ce classes. The swap was seamless because of the design structure.

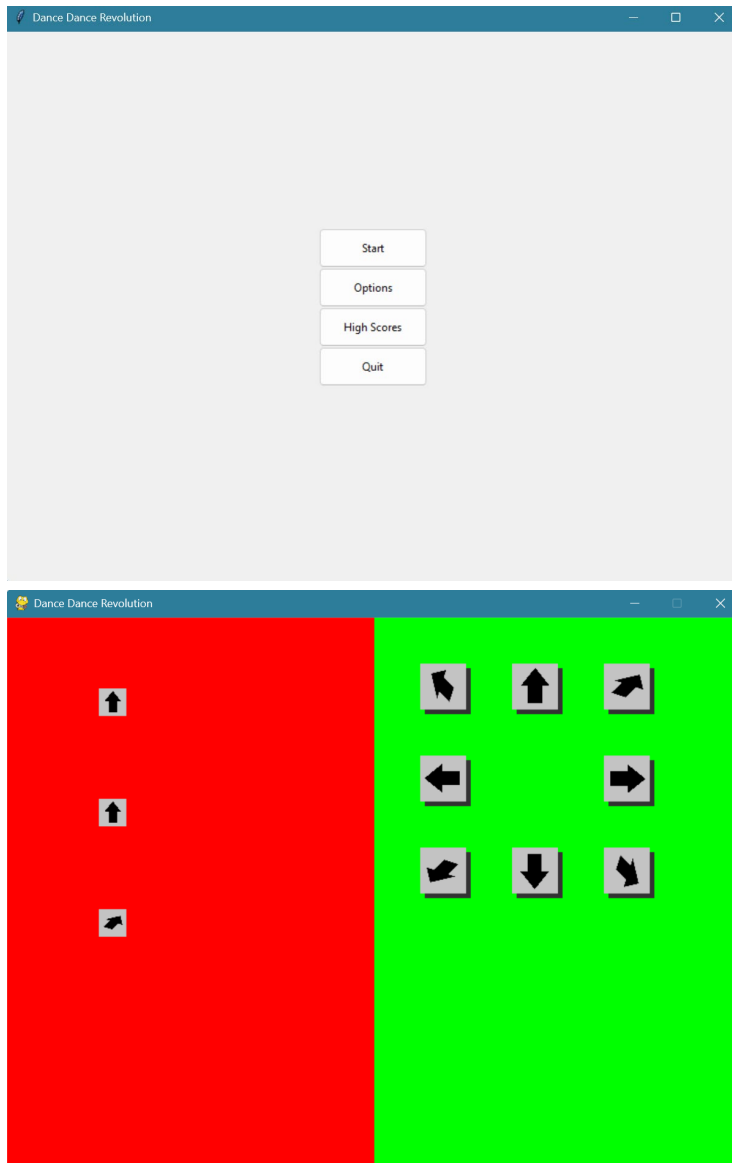
I use a button hanlder to contain all the logic for the buttons instead of trying to spread the buttons out across different classes. This allowed me to fix some issues with the

I use seperation of concerns. Each function only deals with one aspect of the code. Each class focuses on a clearly defined aspect of the code.

The factory pattern is used in left_screen.py for creating arrows to add new arrows to the scrolling screen.

The game is a poartial success from what we were hoping to implement with the stage one documentation. The feel of the game is unpolished and there is no music playing yet. Overall though, the design patters and class structure are there and the project is in a mostly deliverable state.

UnitTesting



```
C:\Users\adamj\OneDrive\Documents\College\SUU\Fall2025\CS-2450\dance_dance_revolution\src\unittests
..searching for item: ' TestLength: ' got 1234
searching for item: ' TestWords: ' got Correct test words
An error occurred: Keyword cannot be empty or whitespace.
searching for item: ' ' got <class 'ValueError'>
Warning: The keyword ' ' was not found in testFile.txt.
A warning occurred: Keyword not found.
searching for item: ' ' got None
Warning: The keyword 'TestWord_DoesNotExist' was not found in testFile.txt.
A warning occurred: Keyword not found.
searching for item: ' TestWord_DoesNotExist ' got None
..Testing if path exists: C:\Users\adamj\OneDrive\Documents\College\SUU\Fall2025\CS-2450\dance_dance_revolution\src\unittests
Oh fiddlessticks! an Error Occured: [Errno 13] Permission denied: 'C:\\Users\\adamj\\OneDrive\\Documents\\College\\SUU\\Fall2025\\CS-2450\\dance_dance_revolution\\src'
Oh fiddlessticks! an Error Occured: [Errno 13] Permission denied: 'C:\\Users\\adamj\\OneDrive\\Documents\\College\\SUU\\Fall2025\\CS-2450\\dance_dance_revolution\\src\\ '
Oh fiddlessticks! an Error Occured: [Errno 2] No such file or directory: 'C:\\Users\\adamj\\OneDrive\\Documents\\College\\SUU\\Fall2025\\CS-2450\\dance_dance_revolution\\src\\randomFile.txt'
.
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Ran 5 tests in 0.005s

OK
PS C:\Users\adamj\OneDrive\Documents\College\SUU\Fall2025\CS-2450\dance_dance_revolution>
```